Product Requirements Document: Premier League Transfer Value & Bid Predictor

Version: 1.1 (draft) | **Owner:** Peter | **Status:** Pre-build | **Date:** 21 Oct 2026

1. Overview

A web application that estimates two things for any Premier League player:

1. **Market value:** what the player is worth as an asset, based on performance, age, position, contract and club context.
2. **Suggested bid range:** what a buying club should realistically pay *given the transfer situation*. This differs from market value when contract length, club finances or buyer/seller dynamics change the price.

The motivating example: Kylian Mbappé was valued at 160M+ for years, yet Real Madrid signed him on a free transfer because his contract had expired. Market value alone could not explain that outcome. Bid value can.

The app explains every prediction (SHAP-based) and surfaces over- and under-valued players through a value-gap leaderboard.

2. Problem Statement

Fans, student analysts and football-analytics hobbyists can see Transfermarkt valuations, but those numbers do not tell them (a) why a player is valued that way, or (b) what a deal would actually cost. Fees routinely diverge from valuations because of contract status, selling-club pressure, release clauses and buyer wealth. No simple public tool models both sides and explains the difference.

3. Goals and Non-Goals

Goals

- Predict current market value for Premier League players with error clearly better than a simple age + position + league baseline.
- Predict a bid ratio (fee ÷ market value) and combine it with market value to output a **bid range**, not a single number.
- Explain each prediction in plain language with SHAP feature contributions.
- Let users see how changing inputs (especially contract months remaining) changes the bid range.
- Rank players by gap between predicted value and Transfermarkt value, and between bid and market value.

Non-Goals (v1)

- Leagues outside the Premier League.
- Real-time or live-match data.
- Predicting whether a transfer will happen, or which club a player will join.
- Modeling wages, agent fees, or full financial structuring of deals.

- User accounts, saved watchlists, or payments.
- Presenting outputs as official valuations or financial advice.

4. Target Users

User	Need
Football fans	Understand why a player is valued as he is and what a fair fee looks like
Student analysts / portfolio reviewers	A credible, explainable ML project with clear methodology
Fantasy / scouting hobbyists	Spot over- and under-valued players quickly

Primary user for v1: the football-curious analyst who wants numbers *and* reasons.

5. Key User Stories

1. As a user, I search for a player and see his predicted market value next to Transfermarkt's value.
2. As a user, I see a suggested bid range (low / mid / high) for that player.
3. As a user, I see which features pushed the value up or down, with plain-language labels.
4. As a user, I adjust contract months remaining with a slider and watch the bid range change.
5. As a user, I browse a leaderboard of the biggest gaps between model value and Transfermarkt value.
6. As a user, I filter by position, club, age range and contract status.
7. As a user, I read a short methodology page explaining data sources, limits and uncertainty.

6. Functional Requirements

6.1 Market Value Model (Model A)

- **FR-A1:** Predict market value (EUR) per player-season from performance, age, position, contract, club and league features.
- **FR-A2:** Normalize performance stats per 90 minutes and relative to position group.
- **FR-A3:** Train per position group (GK / DEF / MID / FWD) or with position-interacted features; document the choice.
- **FR-A4:** Use a time-based split. Train on earlier seasons and test on the most recent complete season.
- **FR-A5:** Compare against a baseline (age + position + league regression) and a gradient boosting model (XGBoost or LightGBM). Report both.
- **FR-A6:** Prevent leakage: features for a given date may only use data available before that date.

6.2 Bid Value Model (Model B)

- **FR-B1:** Build a training set from historical transfers, pairing each fee with the player's market value at the time of the deal.
- **FR-B2:** Predict the **bid ratio** (fee ÷ market value at transfer time).

- **FR-B3:** Handle free transfers explicitly, either as a separate "free or not" classifier followed by a ratio model for paid deals, or an equivalent documented approach.
- **FR-B4:** Include contract months remaining at time of transfer as a core feature.
- **FR-B5:** Where available, include selling-club financial pressure, buying-club spending power, window timing, and transfer year.
- **FR-B6:** Output **bid range** = predicted market value × ratio interval (e.g. 10th, 50th, 90th percentile or a conformal interval).
- **FR-B7:** Normalize or control for transfer-year inflation.
- **FR-B8:** Flag players whose bid prediction has low confidence (sparse comparable transfers).

6.3 Explainability

- **FR-E1:** SHAP values for both models, shown per player as a ranked list of top contributing features.
- **FR-E2:** Feature names displayed in plain language (e.g. "Contract: 8 months left" rather than a raw column name).
- **FR-E3:** Global feature importance view per model on the methodology page.

6.4 Web Application

- **FR-W1:** Player search with autocomplete.
- **FR-W2:** Player page: predicted market value, Transfermarkt value, bid range, SHAP breakdown, key stats.
- **FR-W3:** Contract slider on the player page that re-runs Model B in real time and updates the bid range.
- **FR-W4:** Value-gap leaderboard (sortable, filterable) covering both gap types.
- **FR-W5:** Methodology and limitations page.
- **FR-W6:** A visible disclaimer that figures are model estimates and not official valuations or advice.

6.5 Data Pipeline

- **FR-D1:** Ingest performance data (FBref, Understat), market values and transfer history (Transfermarkt dataset or an equivalent).
- **FR-D2:** Maintain a player ID cross-reference table so records match across sources.
- **FR-D3:** Cache raw data locally as versioned snapshots; the app must not scrape live at request time.
- **FR-D4:** Log and document exclusions (loans, youth players with no history, missing contract data).
- **FR-D5:** Reproducible pipeline: one command rebuilds the processed dataset from raw snapshots.

7. Data Requirements

Data	Source (candidate)	Notes
Player performance stats	FBref, Understat	Multiple seasons; per-90 normalization
Market values over time	Transfermarkt dataset (Kaggle / GitHub)	player_valuations table gives dated value history, used for as-of-date features

Data	Source (candidate)	Notes
Transfer history and fees	Transfermarkt dataset	transfers table includes fee and the player's market value at transfer time; keep free transfers and flag loans
Contract expiry dates	Transfermarkt dataset	Only current expiry is available; contract length at transfer time must be reconstructed
Club context	Derived	League position, spending, financial pressure proxies

Key data risks: name matching across sources, licensing and terms of service for scraping, missing release-clause and bidding-war information, the small number of relevant transfers (hundreds, not thousands), and the Transfermarkt dataset's update freeze (see section 14).

8. Success Metrics

Model metrics

Metric	Target (initial)
Model A: median absolute percentage error on holdout season	Meaningfully below the baseline; stretch goal under 25%
Model A: R^2 on log(value)	Above baseline by a clear margin
Model B: fraction of actual fees inside the predicted bid range	Close to the nominal interval coverage (e.g. around 80% for an 80% interval)
Model B: free-transfer classification	Report precision and recall; contract-expiry cases should be caught reliably

Targets are provisional and should be revised once baseline results are known.

Product metrics (if shared publicly)

- Player pages viewed per session.
- Contract slider usage rate.
- Leaderboard visits.

9. Assumptions and Constraints

- Transfermarkt market value is the primary target for Model A; it is itself an estimate, not ground truth.
- Bid ratio data is sparse and noisy, so ranges are shown instead of point estimates.
- The Transfermarkt dataset snapshot is current only to early July 2026 (valuations to 12 June 2026). v1 trains on historical seasons and treats that snapshot as the "latest" data.
- A free-tier hosting target (e.g. Streamlit Cloud, Render or Hugging Face Spaces) with pre-computed or lightweight inference.
- Development will be done with AI coding agents under a phase-gated process, with a verified checkpoint before each phase advances.
- Data usage must respect each source's terms; prefer existing datasets where scraping is questionable.

10. Risks and Mitigations

Risk	Impact	Mitigation
Data leakage (post-transfer stats used for pre-transfer prediction)	Inflated, misleading accuracy	Strict as-of-date feature construction; time-based split
Small bid-model dataset	High variance	Predict ratio, not raw fee; show ranges; flag low-confidence cases
Name and ID mismatches across sources	Dropped or wrong rows	Cross-reference table, fuzzy matching with manual review of edge cases
Scraping blocked or against terms	No data	Use existing datasets and cached snapshots
Dataset updates paused	No 2026/27 values	Train on history; state the data cut-off in the app; watch for updates or add a carefully rate-limited scraper
Missing context (release clauses, player demands)	Bid range misses real-world fee	Clear disclaimer; frame output as a *fair* bid, not a forecast
Fee inflation over time	Old transfers mislead the model	Year feature or inflation normalization
Users over-trust the numbers	Misuse as advice	Disclaimers and a visible uncertainty display

11. Delivery Phases

Each phase has a checkpoint that must be verified before the next begins.

Phase	Scope	Checkpoint
0. Setup	Repo, environment, rules files, data-licensing decisions	Environment runs; sources and licenses decided
1. Data collection	Ingest stats, values, transfers; build ID cross-reference	Raw snapshots saved; match rate reported
2. Cleaning and features	As-of-date features, per-90 and position normalization	Leakage checks pass; feature table documented
3. Model A	Baseline and gradient boosting for market value	Beats baseline on holdout season
4. Model B	Transfer dataset, free-transfer handling, ratio model, intervals	Interval coverage close to nominal; free cases handled
5. Explainability	SHAP for both models, plain-language labels	Explanations render correctly for sample players
6. Web app	Search, player page, contract slider, leaderboard, methodology	All user stories demonstrable end to end
7. Polish and deploy	Disclaimers, performance, deployment, README	Public URL works; README reproduces the pipeline

12. Open Questions

1 Which Transfermarkt access route will be used (the pre-built dataset only, or also a scraper for data after the freeze)?

2. Should the player universe include only current Premier League players, or also past players for training?
3. Separate models per position group, or one model with interactions?
4. How far back should training data go, and how will older fees be adjusted?
5. Which interval method for the bid range: quantile regression, conformal prediction, or residual-based?
6. Is real-time slider inference needed, or is a pre-computed grid enough for v1?

13. Out of Scope for Later Versions

- Other leagues and cross-league transfers.
- Wage and total-cost-of-signing modeling.
- Buyer-fit recommendations (which club should buy which player).
- Injury-risk modeling.
- Saved watchlists and user accounts.

14. Resource Acquisition Plan

This section lists everything the project needs and exactly how to get it. All items below can be obtained for free. Items marked **Manual** must be done by you personally (accounts, terms checks, tokens); items marked **Agent** can be delegated to the AI coding agent once the manual steps are done.

14.1 Data Resources

Resource	Used for	How to acquire	Owner	Fallback
Transfermarkt dataset (player profiles, valuation history, transfers and fees, clubs, appearances)	Targets for Models A and B; contract features	Download "Football Data from Transfermarkt" by davidcariboo from Kaggle (davidcariboo/player-scores), or the DuckDB/zip file linked from the GitHub repo dcaribou/transfermarkt-datasets. License shown on the repo: CC0-1.0.	Manual (download), Agent (load and clean)	A rate-limited community scraper, after reading the site's terms
Player performance stats (passing, defensive actions, dribbles)	Performance features	Python package soccerdata (FBref) or ScraperFC. FBref enforces strict rate limits, so the agent must add delays and cache every response.	Agent	Dataset appearances table (goals, assists, minutes only)
Expected goals and assists	Attacking quality features	Understat via the understat package or soccerdata.	Agent	FBref xG tables, if available
Contract length at time of transfer	Core feature for Model B	The dataset only has current contract expiry. Reconstruct where possible and spot-check 20 to 30 well-known deals by hand. Drop transfers where it cannot be established and record how many.	Agent, Manual (spot-check)	"Contract status unknown" flag

Resource	Used for	How to acquire	Owner	Fallback
Club financial context	Selling and buying club features	Start with proxies from the transfer data (net spend, league finish). Optional: public wage and revenue figures entered by hand into a CSV.	Agent / Manual	Proxies only in v1
Release clauses, player demands, bidding wars	Would improve Model B	Not available in any structured public dataset. Treat as unobservable and say so in the app.	n/a	n/a

Dataset schema highlights. players: player_id, name, position, sub_position, foot, height_in_cm, date_of_birth, country_of_citizenship, current_club_id, contract_expiration_date, market_value_in_eur, highest_market_value_in_eur. player_valuations: player_id, date, market_value_in_eur, current_club_id. transfers: player_id, transfer_date, transfer_season, from_club_id, to_club_id, transfer_fee, market_value_in_eur (market value at the time of transfer). appearances: player_id, game_id, date, goals, assists, minutes_played, yellow_cards, red_cards. Exact column names should be confirmed after download.

Data freeze. The dataset's updates are paused and it is current to 6 July 2026; player_valuations has nothing after 12 June 2026 and 2026/27 squads are not covered. It is fine for training on past seasons. Check the repository's status announcement for any resumed updates.

Before relying on FBref: confirm which advanced metrics it currently offers, because its data-provider arrangements have changed in the past. Do this in Phase 0 by manually opening a few player pages.

14.2 Accounts and Access

Resource	How to acquire	Owner
GitHub account and private repository	Create the repo, add a README, and commit the rules files (PROJECT.md, RULES.md, IMPLEMENTATION_PLAN.md) first.	Manual
Kaggle account and API token (optional if you use the GitHub download)	Sign up, open Account settings, choose "Create New Token", save kaggle.json to the folder Kaggle's CLI expects. Never commit this file.	Manual
AI coding agent access	Ensure your account and usage limits cover a multi-day build. Give it the PRD and rules files as context.	Manual
Hosting account	Free account on Streamlit Community Cloud, Hugging Face Spaces or Render, connected to GitHub.	Manual

14.3 Software and Libraries (all free)

- **Python 3.11 or newer** with a virtual environment manager (venv, uv or poetry).
- **Data:** pandas, pyarrow, duckdb (optional), soccerdata, understat, rapidfuzz (name matching).
- **Modeling:** scikit-learn, xgboost or lightgbm, shap, and a conformal or quantile tool such as MAPIE for bid-range intervals.
- **App:** Streamlit for the fastest v1, or FastAPI plus a small front end.
- **Quality:** pytest, ruff or black, and a pinned requirements file.

The agent installs these. Your job is only to approve the dependency list in Phase 0.

14.4 Compute and Storage

- Training gradient boosting on tens of thousands of rows runs comfortably on a laptop CPU. No GPU or paid cloud compute is needed.
- Raw data snapshots are likely tens to a few hundred megabytes. Keep them out of Git and commit only small sample files for tests.
- Store processed data as Parquet files and trained models as versioned files.

14.5 Legal and Licensing Checks (Manual)

1. Read the Transfermarkt terms of use and decide whether you will rely only on the pre-built dataset or also scrape.
2. Confirm the license on every dataset you use and note any attribution requirement in the README.
3. Read FBref and Understat terms on automated access and redistribution. Do not publish raw scraped tables in the app; publish only derived predictions and aggregates.
4. Keep a short DATA_SOURCES.md in the repo listing each source, date retrieved, license and any restrictions.

14.6 Order of Acquisition

Step	Action	Done when
1	Create GitHub repo, hosting account (and Kaggle token if needed)	Each account works and tokens are stored safely
2	Complete the legal and licensing checks (14.5)	DATA_SOURCES.md exists
3	Download the Transfermarkt dataset and open the files yourself	You have confirmed values, transfers, contracts and Premier League clubs are present
4	Manually sample 5 to 10 players across FBref and Understat	Needed fields confirmed present
5	Agent pulls and caches FBref and Understat stats with delays	Raw snapshots saved and dated
6	Agent builds the player ID cross-reference; you review the unmatched list	Match rate reported and edge cases resolved
7	Agent reconstructs contract-at-transfer; you spot-check known deals	Spot-check passes and dropped counts recorded

Only after step 7 should Phase 2 (cleaning and features) begin.

14.7 Resource Risks

Risk	Mitigation
A source blocks automated access	Rely on cached snapshots and the pre-built dataset; never scrape at request time
A source changes its columns or removes data	Pin snapshot dates; verify fields in Phase 0 before designing features around them
Dataset updates stay paused	Train on history; show the data cut-off date in the app; consider a careful scraper for recent data

Risk	Mitigation
Historical contract data is incomplete	Spot-check, flag unknowns, and report how many transfers were usable
Too few usable transfers for Model B	Widen the training universe beyond the Premier League, while keeping the app Premier League only